# α-$BaCu_2Se_2$ Thin-Film XRD Analysis and Bayesian Optimization

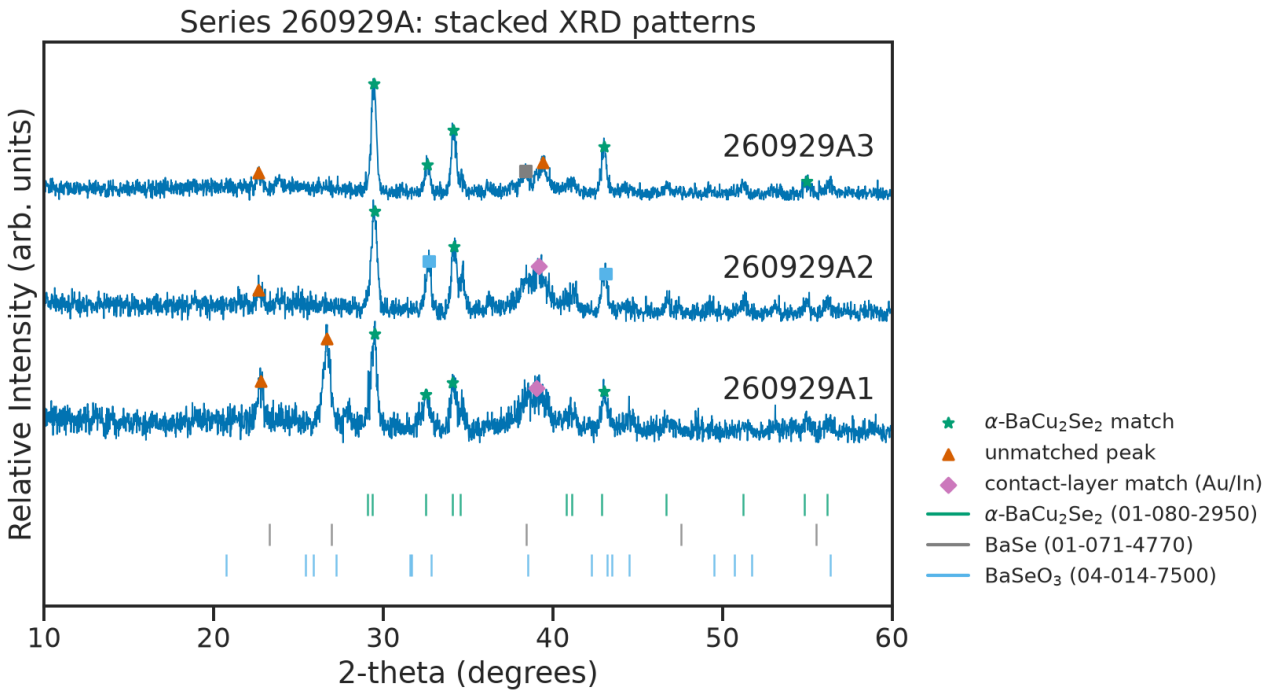
*NOMAD-driven data extraction, phase matching, and GP-based synthesis planning*

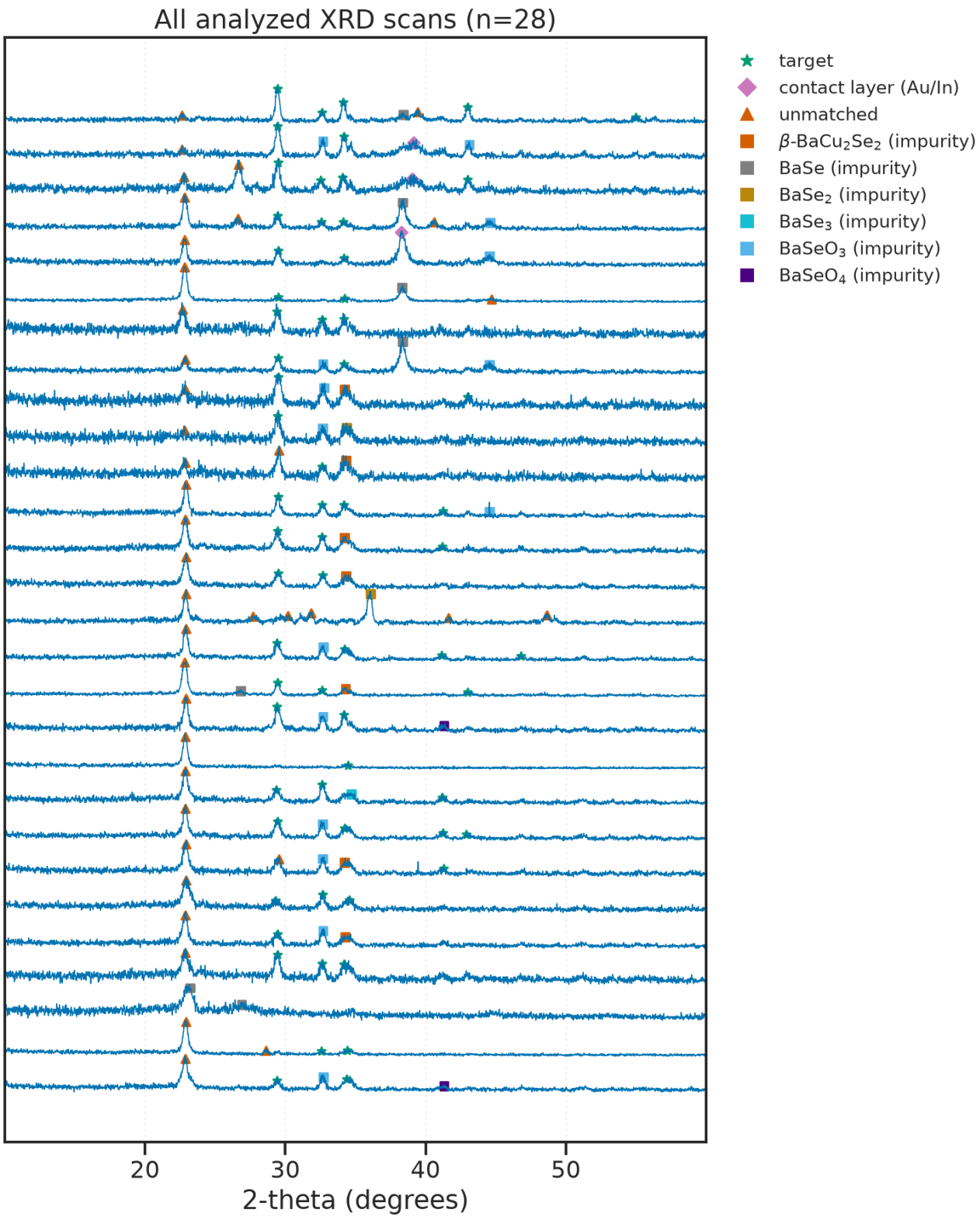
## 1. What was done

Synthesis records and XRD scans were pulled directly from the NOMAD Oasis REST API for the 42 α-$BaCu_2Se_2$ thin-film samples (INLThinFilmStack entries). Synthesis parameters (spin coating, soft/hard bake, selenization) and selenium-pulse settings were extracted for all samples; 35 include pulse on/off data. Each record also carries the precursor Cu/(Ba+Cu) ratio, the number of spin-coating cycles (deposited layers, excluding dynamic-dispense/seed-layer preparation steps), the Mo-coating flag, and the spin-to-selenization day gap. For the 28 films with XRD scans, diffraction patterns were matched against reference cards for the α-$BaCu_2Se_2$ target (01-080-2950) and the impurities β-$BaCu_2Se_2$ (04-008-2861), $Cu_2Se$ (01-080-4377), CuSe (00-049-1456), BaSe (01-071-4770), $BaSe_2$ (04-004-8560), $BaSe_3$ (04-004-8559), $BaSeO_3$ (04-014-7500) and $BaSeO_4$ (04-011-7418); the α target always wins an overlap, otherwise the closest reflection decides. Au (00-004-0784) and In (00-005-0642) are classified separately as contact-layer matches with one shared plot symbol; their detected-peak intervals are excluded from phase-purity accounting and the unmatched/overlap penalty. In matching is restricted to series 260929A. Phase purity and crystallinity (1/FWHM of the strongest target match) were computed per film. A Gaussian-process Bayesian optimizer was trained on the 24 complete films to recommend the next synthesis conditions, optimizing objective = (0.7 × phase purity + 0.3 × normalized crystallinity) × (1 – 0.25 × A) over a five-input space: hard-bake temperature, selenization temperature and time, Cu/(Ba+Cu) ratio, and continuous Se pulse duty cycle. The on+off period is fixed at 3200 ms (3.2 s), so pulse frequency is fixed at 0.3125 Hz and is reported as metadata, not optimized. Intermediate duty cycles are model interpolations from three observed levels and should be treated as exploratory; recommended on-times are quantized to 1 ms increments. A long-form CSV of classified detected peaks is exported for sample-level audit. CBC is defined as $n_{Cu}/(n_{Ba}+n_{Cu})$; the candidate grid is 0.63–0.71, so a ratio of 1 (no Ba) is excluded. A is the unmatched/overlap share of detected peak-interval area, a screening proxy—not a quantitative phase fraction or a confirmed assignment of the ~23° feature.

## 2. XRD phase identification

Stacked XRD patterns per series with reference bars for the α target (always shown) and only those impurity phases matched in the series' patterns. Markers flag target (star), impurity (square, colored by phase), shared Au/In contact (diamond), and unmatched (triangle) peaks. Au/In reference-card bars are omitted.

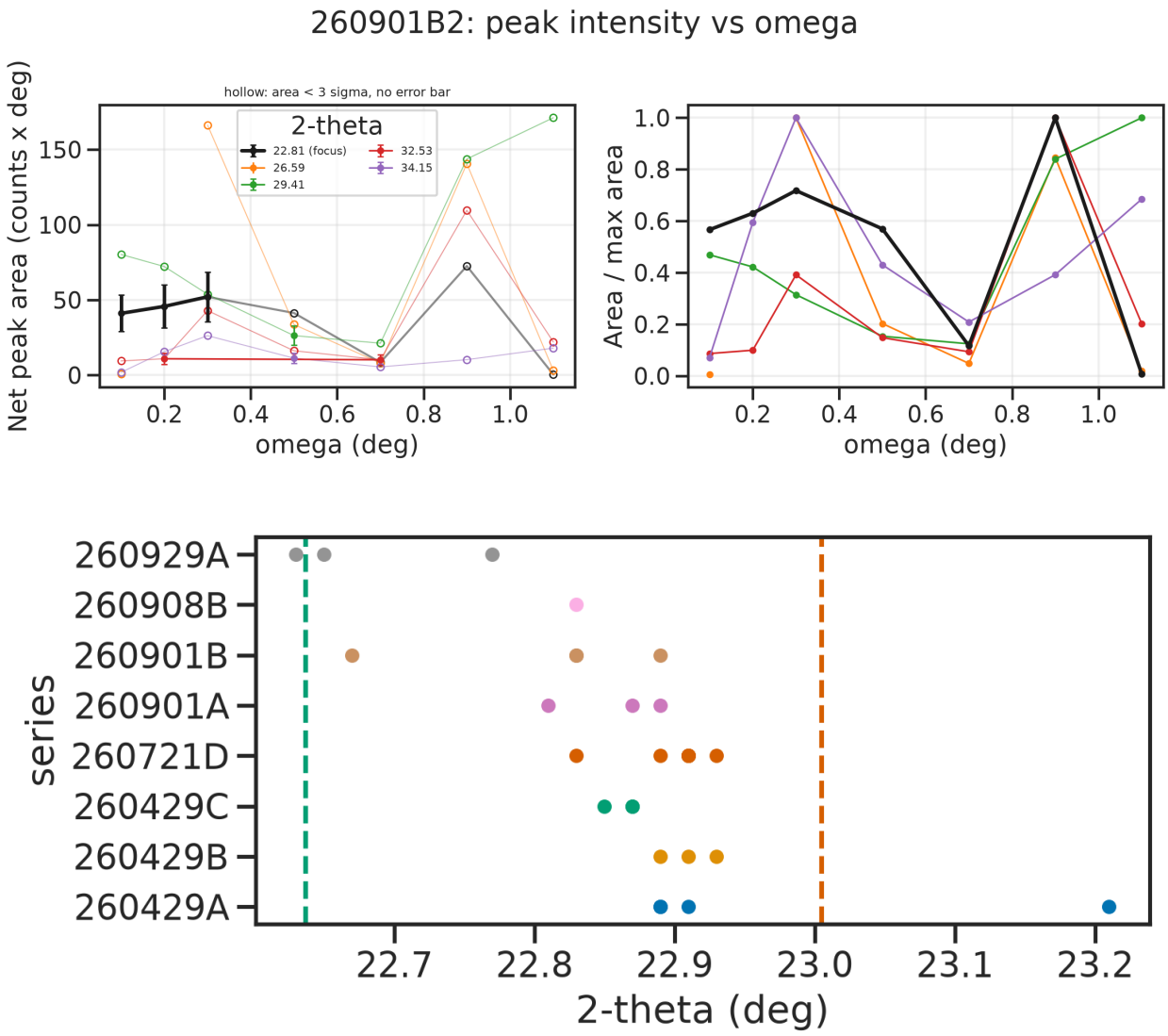

Series 260929A: stacked XRD patterns
Relative Intensity (arb. units)
260929A3
260929A2
260929A1
10
20
30
40
50
60
2-theta (degrees)
α-BaCu2Se2 match
unmatched peak
contact-layer match (Au/In)
α-BaCu2Se2 (01-080-2950)
BaSe (01-071-4770)
BaSeO3 (04-014-7500)



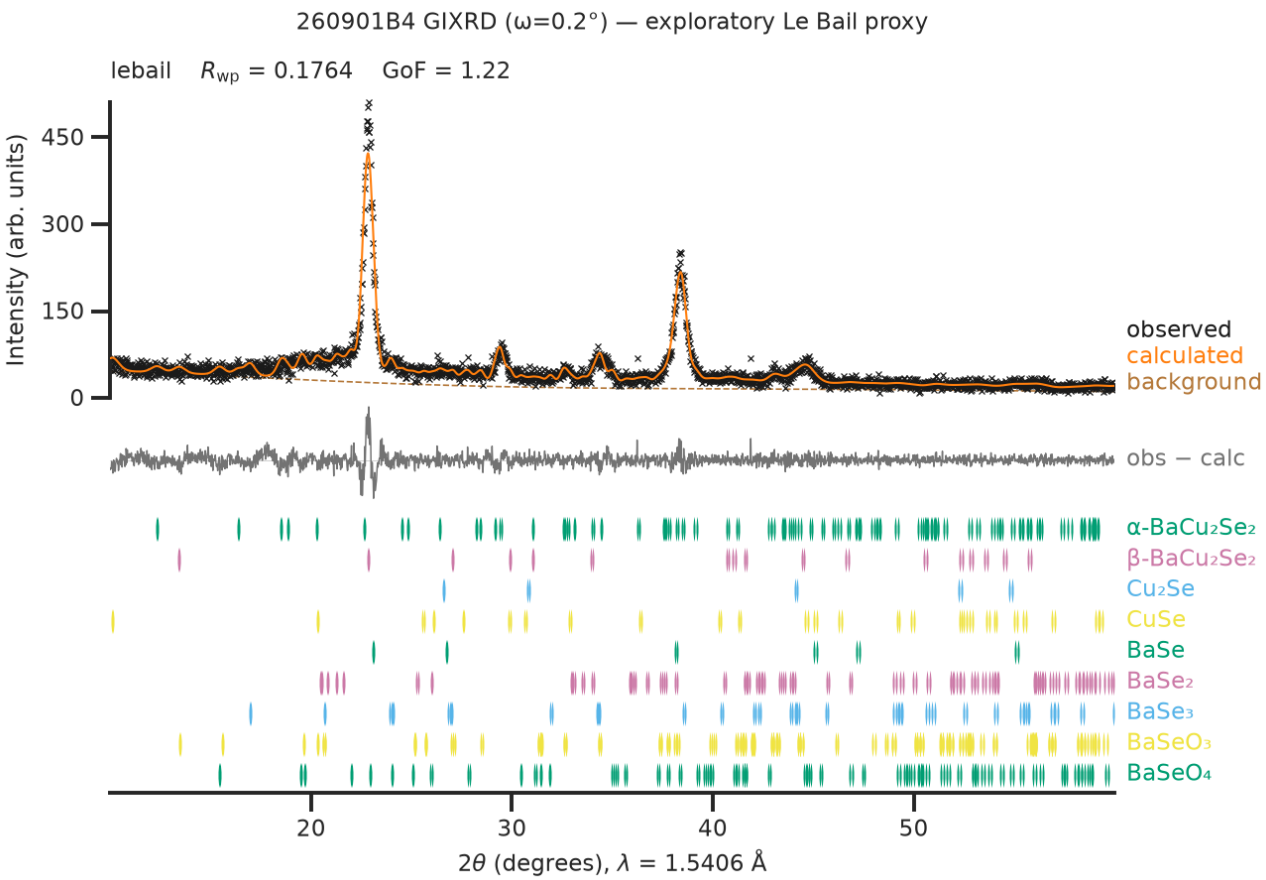

## Omega study of sample 260901B2

Seven grazing-incidence scans (omega 0.1–1.1°, 20–35° 2θ) were fitted peak by peak to test the origin of the ~23° feature (notebook 2.1; this sample only, not used by the optimizer). The feature sits at 22.81° 2θ with FWHM ≈ 0.4–0.5°. The 10 %-intensity matching used for purity does not assign it, but the cards list weak lines near it: α-$BaCu_2Se_2$ at 22.64° (4.6 %) and β-$BaCu_2Se_2$ at 23.01° (3.0 %), 0.17° below and 0.20° above the sample 260901B2 peak. Across all films the feature sits at 22.86–22.98° in series 260429A/B/C, 260721D and 260901A (closer to the β line) and at 22.68° in series 260929A (closer to the

α line), so it is a film-side reflection of the $BaCu_2Se_2$ family rather than a Cu–Se, Ba–Se or Ba–Se–O impurity line. In the eight films with the strongest peak it is 30–100 times more intense relative to the main α and β lines than the cards predict, so strong texture is required. A second-order peak at 46.6–46.9° ($d_1/d_2$ = 2.00) gives d = 3.87–3.89 Å, closer to β (101) at 3.863 Å than to α (011) at 3.925 Å; the β (103) line at 30.02° is at most marginal. This favours the β polymorph but matches neither card exactly, so the assignment remains unconfirmed; a symmetric θ–2θ scan, a rocking curve and a pole figure at 22.9° would settle it. The SLG substrate shows no sharp peak at 23°.

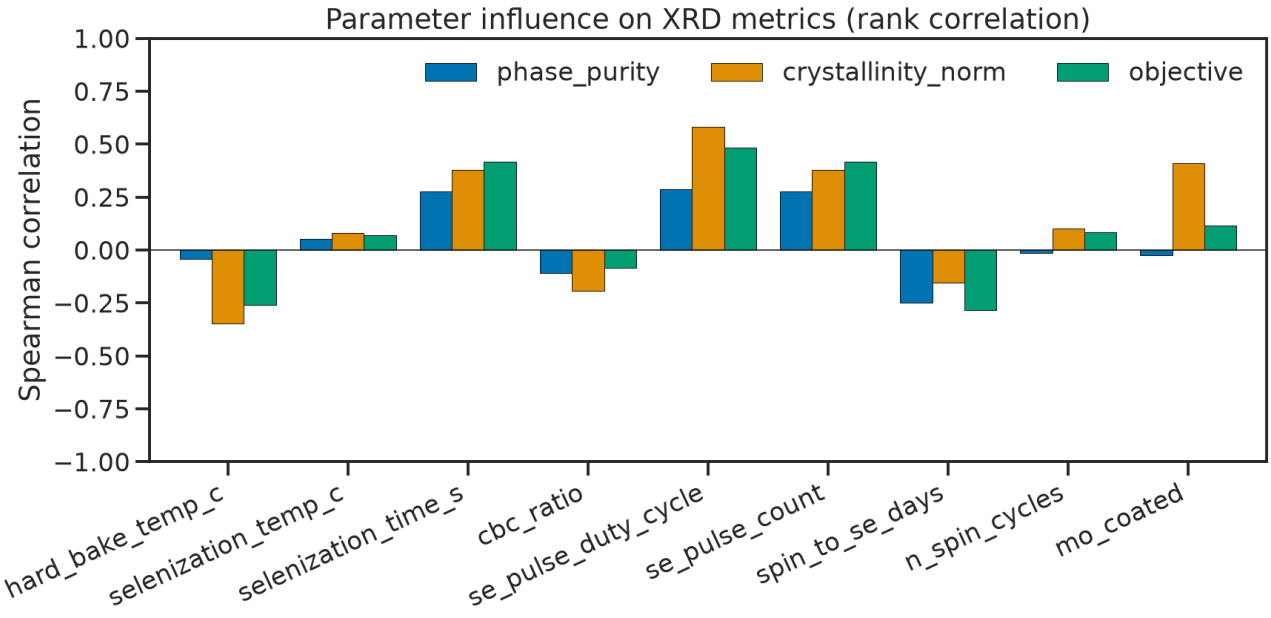


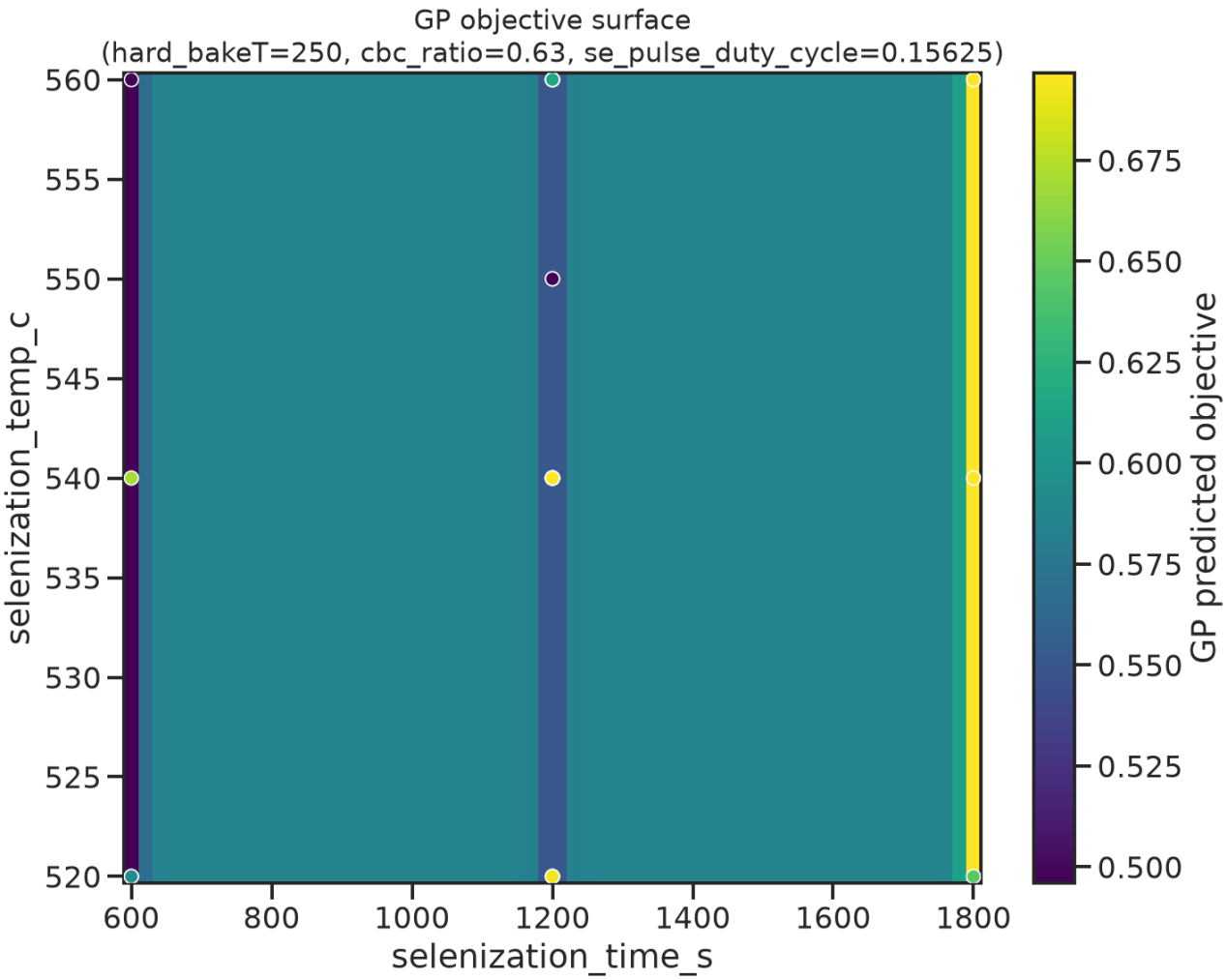

An exploratory Le Bail profile fit (Bragg–Brentano proxy, cells fixed) was repeated for all 28 scans with nine CIF phases (α- and β-$BaCu_2Se_2$, $Cu_2Se$, CuSe, BaSe, $BaSe_2$, $BaSe_3$, $BaSeO_3$, $BaSeO_4$) instead of five. Mean $R_{wp}$ fell from 0.190 to 0.184 and GoF from 1.29 to 1.24. In the 22.1–23.1° window the RMSE fell from 23.0 to 19.9 counts (lower in 26 of 28 scans) and the calculated peak position error from 0.074° to 0.053°, but the calculated peak height is still only 78 % of the observed one and a derivative-shaped residual remains at the peak. Le Bail intensities are free and the cells are fixed, so the better fit shows that the extra phases can account for the profile, not that β rather than α produces the 23° peak.




## 3. Main results

Across the analyzed films the mean phase purity is 0.68 and mean crystallinity is 2.16. The best film under the penalized score is 260429B2 (score 0.88; unpenalized score 1.00; purity 1.00, crystallinity 2.92, A 0.49). The optimizer uses the five selected inputs only. Other process and substrate descriptors remain available for descriptive exploration; duty cycle is modeled continuously, with the fixed pulse frequency excluded from the optimizer because it has no observed variation.



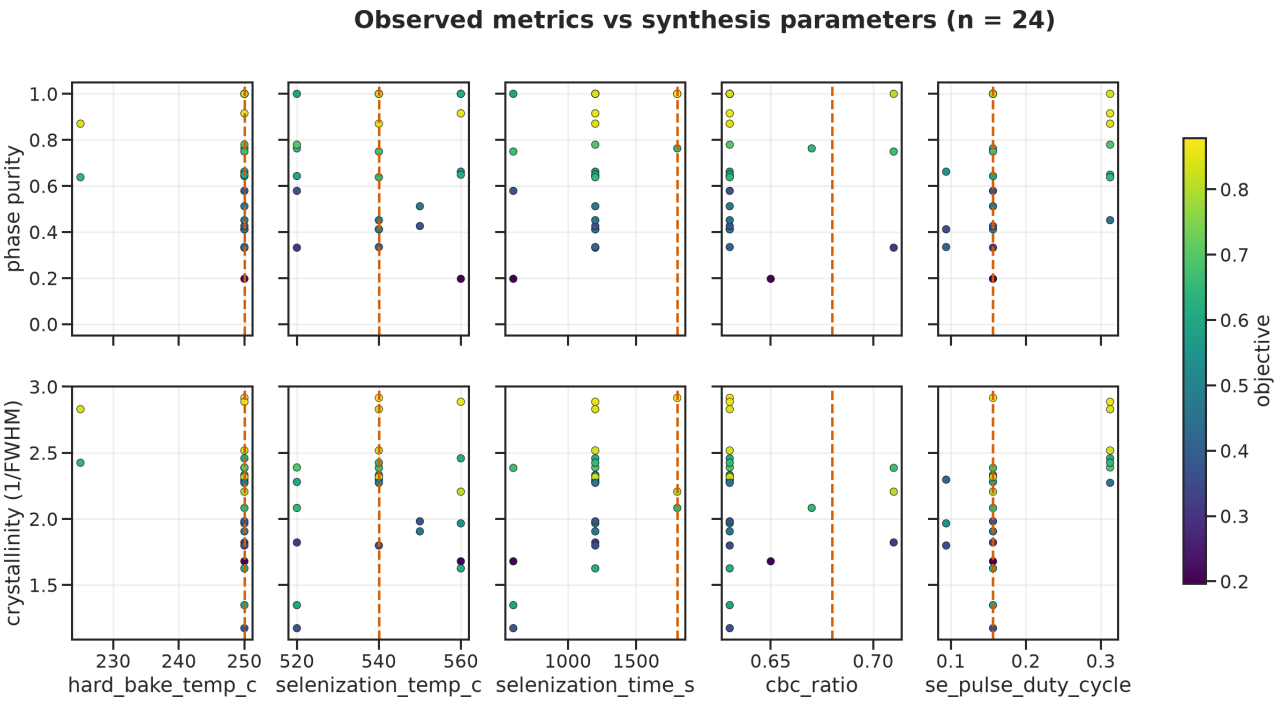



## 4. Suggested next experiments

Top Gaussian-process suggestions (predicted objective ± std), ranked by expected improvement:

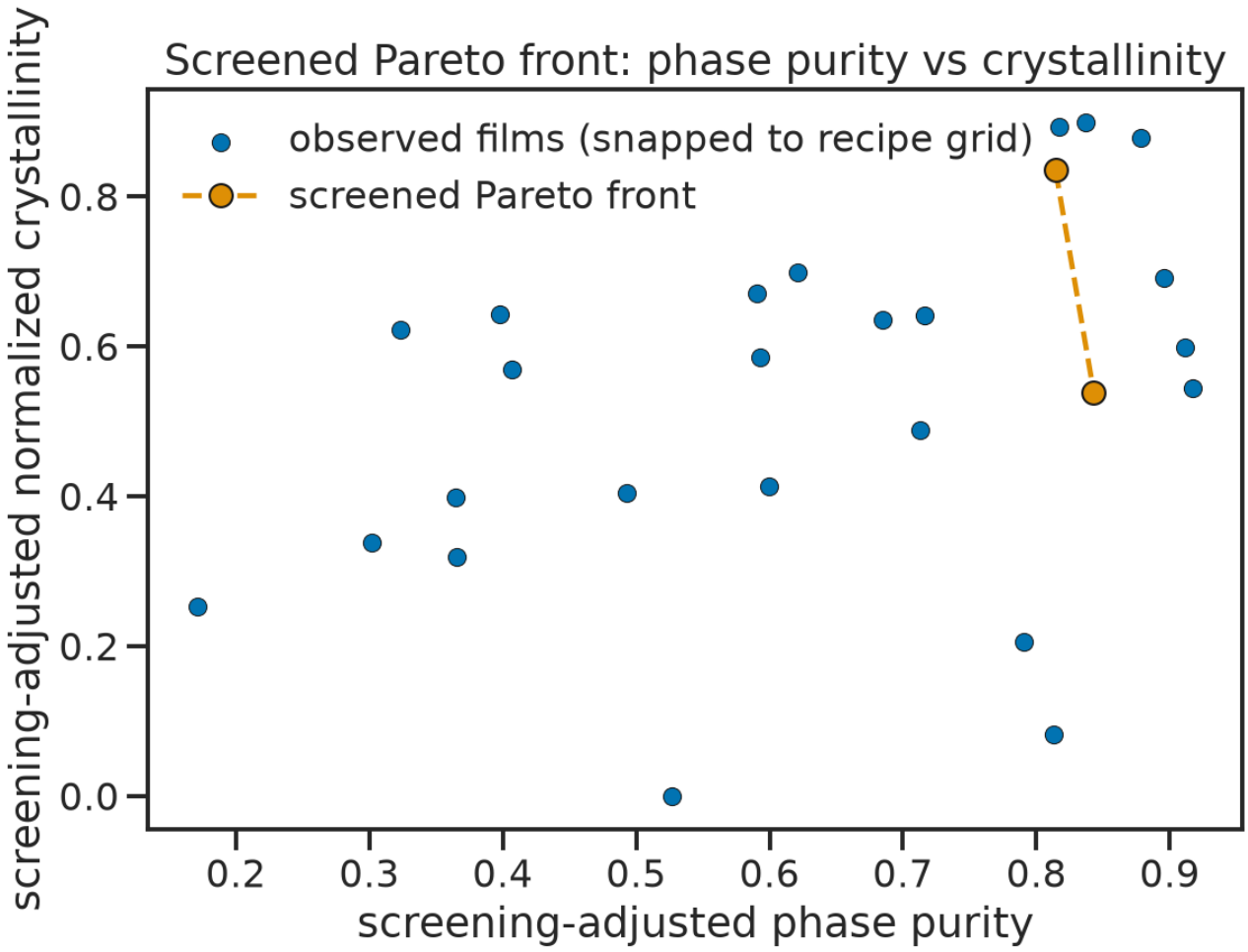
| # | Hard bake °C | Selen. temp °C | Selen. time s | CBC ratio | Se duty cycle | Se on ms | Se off ms | Predicted score | GP std |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 250 | 540 | 1800 | 0.680 | 0.1562 | 500 | 2700 | 0.682 | 0.170 |
| 2 | 225 | 550 | 1800 | 0.630 | 0.3125 | 1000 | 2200 | 0.709 | 0.154 |
| 3 | 250 | 540 | 1800 | 0.660 | 0.2963 | 948 | 2252 | 0.662 | 0.179 |
| 4 | 225 | 540 | 1800 | 0.700 | 0.2931 | 938 | 2262 | 0.663 | 0.179 |
| 5 | 225 | 540 | 1800 | 0.690 | 0.2959 | 947 | 2253 | 0.662 | 0.179 |

Recommended first experiment: hard bake 250 °C, selenization 540 °C for 1800 s, CBC ratio 0.680, and Se pulse duty cycle 0.1562 (500 ms on / 2700 ms off) (predicted objective 0.68).




## 5. Multi-objective optimization (Ax)

A complementary multi-objective search treats screening-adjusted phase purity and normalized crystallinity as two separate objectives: both are multiplied by the same (1 – 0.25 × A) factor. Ax (BoTorch) fits a Gaussian process per metric and finds the Pareto front via qNEHVI hypervolume maximization. Its five candidate inputs match notebook 3: hard-bake temperature, selenization temperature/time, CBC ratio, and continuous pulse duty cycle. Pulse period is fixed at 3.2 s (0.3125 Hz), so frequency is excluded as a constant. Duty-cycle suggestions are quantized to 1 ms on-time increments; interpolation between the three observed duty levels is exploratory. Numeric process recipes use the same explicit experiment-scale grids, including 25 °C bake steps. This remains a Pareto formulation, not the same scalar objective as notebook 3. A is an exploratory detected-peak-area screening proxy, not a phase fraction or a confirmed assignment of the ~23° feature.




| # | Hard bake °C | Selen. temp °C | Selen. time s | CBC ratio | Se duty cycle | Se on ms | Se off ms |
|---|---|---|---|---|---|---|---|
| 1 | 225 | 560 | 1500 | 0.630 | 0.3125 | 1000 | 2200 |
| 2 | 225 | 540 | 1800 | 0.630 | 0.1703 | 545 | 2655 |
| 3 | 250 | 560 | 1800 | 0.640 | 0.3125 | 1000 | 2200 |
| 4 | 250 | 530 | 1500 | 0.630 | 0.1897 | 607 | 2593 |
| 5 | 250 | 560 | 1800 | 0.630 | 0.0997 | 319 | 2881 |